

# Project Proposal

## Patient Medical History Generation using AI



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## **Abstract**

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The rise of artificial intelligence (AI) is transforming healthcare significantly, which revolutionizes healthcare professionals' interactions with patients and documenting medical records. This research/project aims to develop an AI-based medical record summarization system for doctors, that will generate a summary by checking their past medical history generally known as Electronic Health Records (EHR), which are patients' medical records saved in a digital form. The medical record summarization system known as “Patient History using AI (PHAI) combines natural language processing and machine learning algorithms and LLMs to generate short summaries. Utilization of takeover quality metrics will enable reliable summaries of patients' medical histories. These Ai generated summaries will help healthcare professionals save time and attend more patients.

PHAI's abilities will extend further than EHR summarization, empowering the system to create new patient records by asking targeted questions, reducing the review time and effort required for healthcare professionals to gather essential medical information. This method will support the productivity of healthcare providers, speed up decision making and improve the quality of patient care by validating that critical medical records are correctly captured and improve accessibility.

## **Introduction**

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Over the past few years, an influx of AI tools in healthcare have been observed. These tools increase the quality of healthcare and save time. These tools cover healthcare areas such as diagnostics, treatment planning, patient monitoring etc. Huge success has been seen in AI Tools that use chatbots for personalized healthcare advice. Another successful application has been in Image Analysis using AI to detect diseases like cancer and Alzheimer's. The motivation for this research comes from the inherent challenges faced when creating manual EHRs and analyzing them can be time consuming, can have numerous errors and inaccuracies, the aim of this research is to develop a system powered by artificial intelligence (AI) for accurate electronic health records (EHRs) which enables healthcare professionals to summarize patient medical histories and store all EHRs at one place to support healthcare professionals in reducing time taken by traditional EHRs and making them paperless, accurate, reliable information and bridge the gap between the rising demand for healthcare and limited doctors, we will study the usage of generative AI in medical practices, how to diagnose and treat also what are its limitations that comes when implementing the system [1]. The system will use

natural language processing (NLP) techniques like Tokenization, Named Entity Recognition, Part of speech Tagging, Dependency Parsing and Deep Learning models like Gemini, Pegasus and GPT 4 to analyze large amount of patient data and create an error free and well organized summary of their medical history, and if a patient has no medical record previously, it can ask it to complete by asking the targeting questions for accurate summary.

The proposed research project is used to enhance the effectiveness and efficiency of EHR generation, thereby enabling physicians to focus on complex, high-value activities. EHR contains patients detailed medical summary records in a digital format that helps healthcare workers to make decisions about patient's health and give complete medical assistance. In this study we will observe how NLP techniques help patients and physicians to share information seamlessly and potential challenges that arise in advancement of the healthcare environment [2]. We aim to collect and use our own dataset form hospital for training and testing of AI models, we will also be using the publicly available datasets for training and testing of the model, our aim is to helps healthcare professionals to reduce review time and improve overall accuracy while focussing on patient centered care. The current EHR system relies on manual data entry, which can be time-consuming and increases the chances of errors or inaccuracies. Physicians spend most of the time creating, reviewing, and updating medical records to understand patient diseases. However, the present EHR system is not intelligent enough and is not able to summarize the medical records by analyzing past medical history, give recommendations to healthcare professionals, accelerate decision making, and reduce the risk of human error.

## **Problem Statement**

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In our healthcare system, a shortage of healthcare providers compared to the large number of patients makes it challenging for doctors to provide timely, comprehensive care. Additionally, the lack of AI-driven systems to create complete, summarized patient histories leads to split and gaps in medical records, while existing EHR systems vary greatly across regions. To address these issues, I aim to develop an AI-based solution that uses medical records datasets both structured and unstructured data to generate organized, detailed patient summaries. This system will help reduce doctors' documentation time, allowing them to focus more on patient care, improve clinical decision-making, and ultimately enhance the quality-of-care patients receive.

## List of Tentative Research Paper

| Study/Research                                                                                                                                                                                       | Technology                                    | Customized AI Agents for Comprehensive EMR Coverage | Handle to incomplete EMR |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------|--------------------------|
| <b>Generative AI for Transformative Healthcare: A Comprehensive Study of Emerging Models, Applications, Case Studies, and Limitations</b>                                                            | GEN AI, LLMS, NLP, GANs, VAEs                 | ✗                                                   | ✗                        |
| <b>AI-Driven Natural Language Processing in Healthcare: Transforming Patient-Provider Communication</b>                                                                                              | NLP, AI, LLMs                                 | ✗                                                   | ✗                        |
| <b>Applying generative AI with retrieval augmented generation to summarize and extract key clinical information from electronic health records.</b>                                                  | RAG, LLMs, zero-shot prompt engineering       | ✗                                                   | ✗                        |
| <b>Intelligent Clinical Documentation Harnessing Generative AI for Patient-Centric Clinical Note Generation</b>                                                                                      | NLP, LLMs, Automatic Speech Recognition (ASR) | ✗                                                   | ✗                        |
| <b>AI-powered Clinical Decision Support Systems for Precision Medicine: Developing AI-powered systems to provide clinicians with decision support tailored to individual patient characteristics</b> | ML, Data Analytics                            | ✗                                                   | ✗                        |
| <b>Transforming healthcare documentation: harnessing the potential of AI to generate discharge summaries</b>                                                                                         | LLMs, NLP                                     | ✗                                                   | ✗                        |

|                                                                                                        |                                  |   |   |
|--------------------------------------------------------------------------------------------------------|----------------------------------|---|---|
| <b>Code empowers large language models for complex tabular reasoning on electronic health records.</b> | LLMs, AutoGen, LLM2SQL, DIN-SQL  | ✗ | ✗ |
| <b>My Research</b><br><b>Patient Medical History Generation using AI (PHAI)</b>                        | LLMs, NLP, EHR Interfacing tool. | ✓ | ✓ |

Table 1. Previous Research Papers vs My Research.

## Research Gap and Research Question

The literature reflects a broad gap in comprehensive use of AI for generation of medical histories for patients. While the completed studies were focused on the automation of clinical documentation and hence better administrative efficiencies, they often missed aspects of real-time patient data and synthesis of structured and unstructured data into coherent medical histories also their entities according to region. For instance, even though RAG-based methods are studied for summarizing clinical data, few works link them to the real-time update aspects concerning patients. Moreover, though patients can provide their feedback, the facility does not prove to serve with even the finest adjustments for individual medical histories. Finally, the quality and completeness of AI-generated patient medical records summaries are still underexplored; some of them are under development. Their experimental effect on the related clinical choices or decisions is inadequate.

To fill those gaps, the following research questions have been formulated for this study:

1. How to handle incomplete records?
2. How can we generate effective prompts for summarizing large data of patient histories, such as 5-year record datasets?
3. What methods can we develop to verify whether the summary generated is good or bad?
4. How is data protection and confidentiality achieved?

All these will contribute significantly to the field in terms of adding more value to patient-centered care and automating clinical workflows.

## Block Diagram

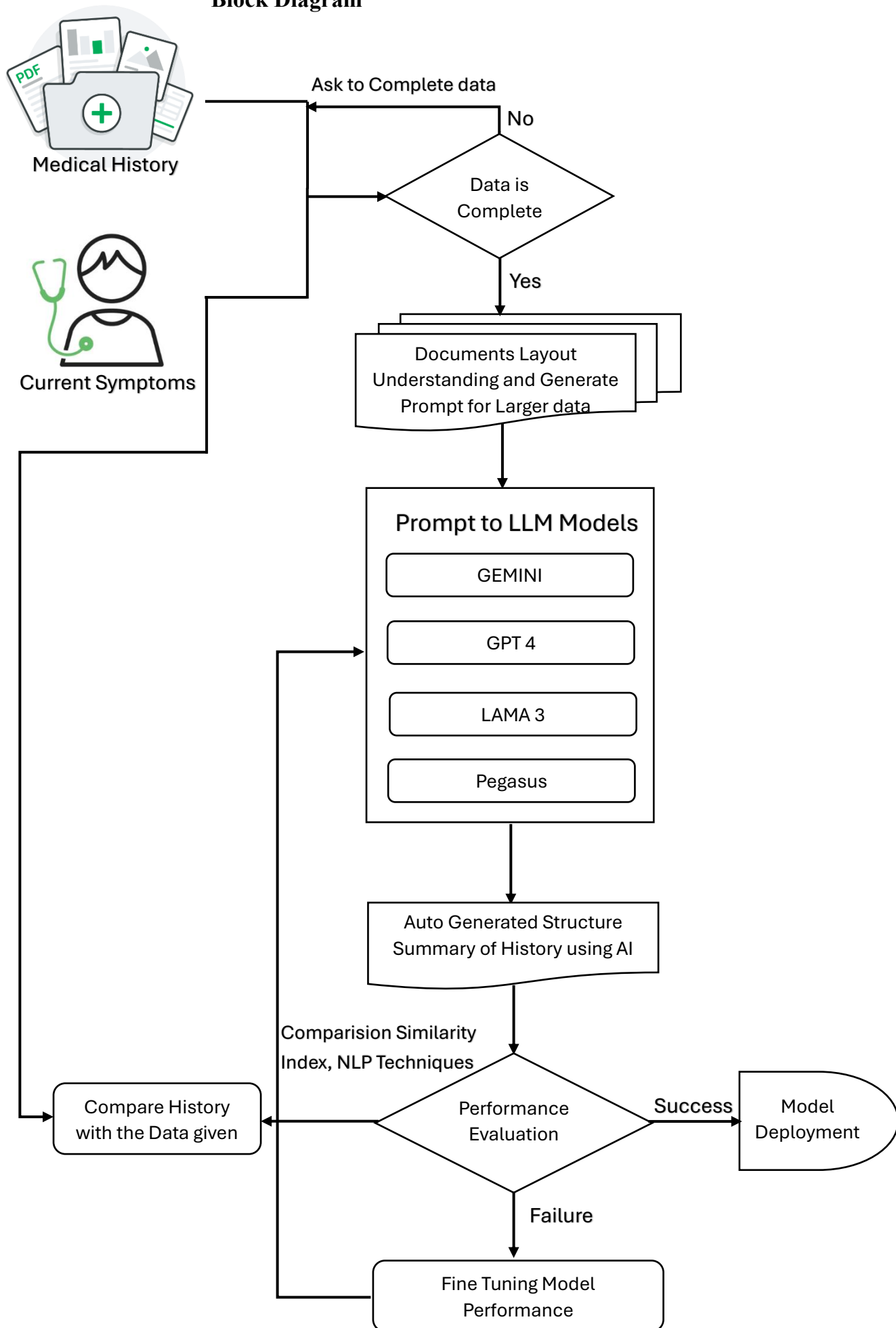


Figure 1. Block Diagram.

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